

Performance Assessment of Artificial Neural Networks-Based MPPT Technique for Photovoltaic Systems

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Abstract—Perturb and Observe (P&O) is one of the common techniques used for Maximum Power Point Tracking (MPPT) of PV panels due to its implementation simplicity. However, this conventional technique suffers from slow peak tracking speed under fast-changing environment conditions. With the revolution of machine learning, Artificial Neural Networks (ANNs) show fast, robust, and time-efficient solutions in different fields that require online data monitoring and fast decision-making. This paper presents a performance assessment of a proposed Artificial Neural Network (ANN) technique for maximum power extraction of Photovoltaic (PV) systems. The proposed technique shows promising solutions for MPPT in fast-changing environments and overcomes the limitations of conventional MPPT techniques.

Keywords— *Maximum Power Point Tracking (MPPT); Photovoltaic Systems (PV); Artificial Neural Networks (ANNs); Perturb & Observe (P&O).*

I. INTRODUCTION

Recently, Photovoltaic (PV) systems show the fastest-growing renewable energy sources due to their ease of use, low operating costs, low noise levels, and low of CO₂ emission [1]. However, low conversion efficiency (around 12% to 25%), high initial costs, and weather dependency limit the widespread installation of PV systems [2]. Therefore, Maximum Power Point Tracking (MPPT) techniques are commonly involved in PV systems to fully utilize the power extraction from PV panels [3]-[7].

Perturb and observe (P&O) method is commonly used for MPPT in PV systems due to its implementation simplicity [8]. P&O technique adjusts the PV modules' terminal voltage continuously then the output power is compared with its previous value during the perturbation cycle. Thus, P&O control technique adapts the PV module operating point with changes occur in the operating voltage to increase the output power of the PV module. However, due to the perturbation and observation cycles, P&O suffers from slow tracking speed finding the peak of the extracted power of the PV module.

Besides, the low dynamics found during tracking the maximum power under various fast change weather conditions [9]-[11].

Machine learning algorithms, specifically artificial neural networks, have shown promising solutions to predict the optimal operating points for PV systems [11]-[14]. In [15], a multiple linear regression (MLR) model has been used to estimate the PV module efficiency by monitoring weather parameters, i.e., radiation and temperature. In [15], ANN-based MPPT technique has been presented to track the maximum power under shading conditions.

In this paper, a novel ANN-based MPPT technique for PV systems is proposed. The proposed technique shows fast and accurate maximum power tracking capability. The main contributions of this work can be summarized as follows:

- A novel ANN-based MPPT technique is proposed.
- Performance assessment of the proposed technique in comparison to conventional P&O MPPT method via MATLAB environment under fast changes in solar radiation and temperature.

The structure of this paper is summarized as follows: Section 2 describes the PV system under investigation; Section 3 overviews the conventional P&O MPPT method; Section 4 presents the proposed ANN-MPPT method; Section 5 addresses the results of the investigation and simulations; Section 6 summarizes and concludes the findings of the presented study.

II. SYSTEM DESCRIPTION

The schematic representation of a PV system connected to a dc variable load is shown in Fig. 1. The main elements of the system can be described as follows:

A. PV Module

The current-voltage characteristics of a real PV cell can be accurately represented by using the single exponential model depicted in Fig. 1. Voltage drop due to the current flowing

through the semiconductor material is represented by a series resistance R_s while leakage current to the ground at the cell limits is represented by a shunt resistance R_{sh} [17], [18].

For modelling the PV module, the cell voltage is divided by the number of series-connected cells (n_s), hence, the equation representing the single exponential model will be [19], [20]:

$$I_{PV} = I_{ph} - I_0 \left(\exp\left(\frac{(V_{PV} + I_{PV} R_s)}{m V_t}\right) - 1 \right) - \left(\frac{V_{PV} + I_{PV} R_s}{R_{sh}} \right) \quad (2)$$

where,

$$V_t = \frac{n_s K T C}{q} \text{ is the panel thermal voltage.}$$

I_{PV} : Output cell current.

V_{PV} : Output cell voltage.

I_{ph} : Photo-generated current.

I_{do} : The saturation current

I_{sh} : The shunt leakage current to ground

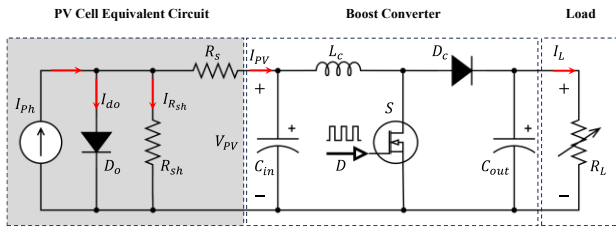


Fig. 1. A schematic representation of a dc load connected to PV panel via a controlled dc-dc boost converter.

B. Boost converter

Boost converter (illustrated in Fig. 1) is one of the common dc-dc converter topologies used to match the PV module voltage to meet the load requirements. Besides, extracting the maximum power from the PV panels [21]. Boost converter functionality is simply explained depending on the semiconductor switch state, i.e., when the semiconductor switch is on, the inductor stores the energy supplied from the PV module. Contrarily, when the switch is off, the stored energy is then transferred to the output. The output capacitor is employed smooth the dc output while the diode is placed to prevent current direction reversal between the output and the input. The semiconductor switching (usually a MOSFET) produces a modulated signal of the output voltage obtained from the MPPT controller. Thus, the output voltage depends on the duty cycle of the switch as in (3) [17], [22].

$$V_{out} = \frac{V_{in}}{(1-D)} \quad (3)$$

III. OVERVIEW OF CONVENTIONAL P&O MPPT METHOD

Perturb and Observe (P&O) method is commonly used in photovoltaic (PV) systems due to its simplicity and low computational requirements. P&O continuously adapts the system's operating point, i.e. the duty cycle, to track the maximum output power. The operating principle can be explained as shown in Fig. 2 and summarized as following: a)

the PV system is set to operate at a certain duty cycle; b) the duty cycle is perturbed, i.e. increased by a small amount; c) the change in the output power is then observed; and d) according to the power difference, the next iteration will be either increase/decrease the duty cycle of the boost converter until the maximum operating point is achieved. Due to the intrinsic behavior of the peak searching method of the P&O, the operating point oscillates around the peak of the P-V curve leading to inefficient energy production [22], [23].

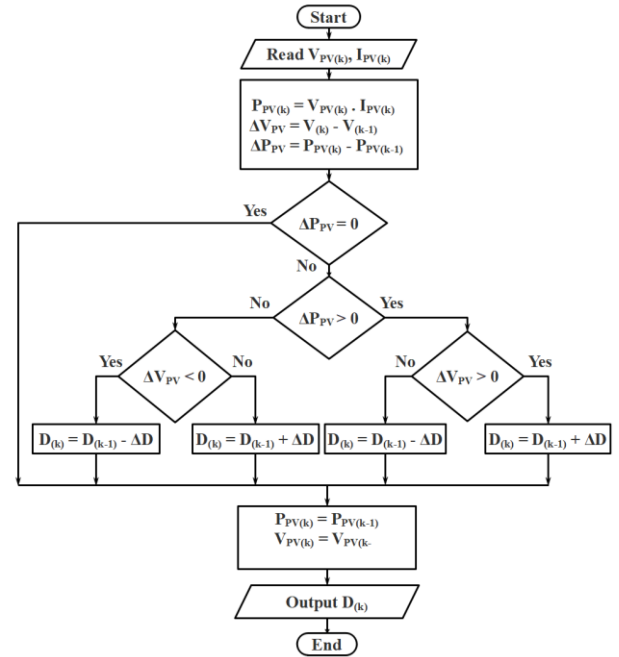


Fig. 2. Flowchart of the conventional Perturb & Observe MPPT method.

IV. PROPOSED ANN-BASED MPPT TECHNIQUE

A. Overview

The ANN is a computer-based model which mimics the human brain's neuron behavior and handles their various functions. The ANN is structured from linked nodes that are interconnected where each network contains three main layers: a) The input layer; b) hidden layer(s); and c) the output layer. These layers handle the processing and the interaction of input and output data. Pattern recognition, clustering, classification, and prediction are the common applications of the ANN due to their adaptivity, ability to self-learning, solve non-linear problems, and the high-speed processing of input-to-output mapping [25].

Feedforward neural networks (FNNs) provide promising solutions due to their capability of solving linear and nonlinear classification problems [41]. The FNN method processes input data to output data without any feedback loops where it can be trained using previously captured data on different environmental variables such as temperature and solar irradiance for MPPT of PV systems. Therefore, the ANN can predict the optimal operating points under different conditions. However, the absence of feedback loops results in a mismatch between target and actual outputs. To avoid this mismatch, the weights and biases of the ANN's linked

nodes are adapted during the training process which is called backpropagation.

The Levenberg-Marquardt (LM) backpropagation algorithm is used in this application for the training and validation of the ANN-based model. The LM algorithm combines two optimization methods: a) the Gradient Descent method, which the sum of the squared errors is minimized by updating the parameters in the steepest-descent direction; and b) the Gauss-Newton method, which the sum of the squared errors is minimized by assuming the least squares function is locally quadratic in the parameters then finding the minimum of this quadratic.

B. Methodology of the Proposed Technique

Due to the nonlinearity and unpredictability nature of PV characteristics, extracting the maximum power becomes challenging. Machine learning, specifically neural networks, offers a promising solution to cope with the limits and disadvantages of conventional MPPT techniques like P&O [25]. The proposed ANN-based MPPT algorithm uses the neural networks features of learning from given predefined data and predictions or decision making with minimal computational effort. The ANN can be trained to accurately predict the MPP from the trained dataset and collate the inputs (such as solar radiation and temperature) and the output duty cycle. The proposed ANN-based MPPT technique can be summarized according to the following stages (see Fig. 3):

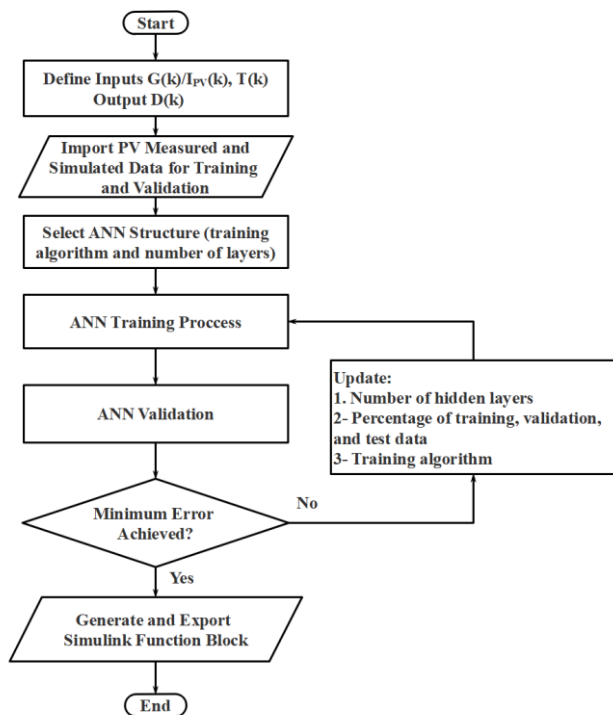


Fig. 3. Flowchart of the proposed ANN-MPPT technique.

1) Data collection and pre-processing

The LG Electronics Inc. LG400N2W-V5 PV module, connected to a variable dc source (see Fig. 4), is employed for the training process of the PV system under different radiation (10-1000 W/m²) and temperature levels (10-70°C).

For each operating condition, the PV module variables such as voltage current and output power are monitored and stored in a data table. Then, a maximum search function is applied to obtain the MPP for each operating condition and stored. The collected data during this process are correlated to the output duty cycle. The data collection and pre-processing tests have been driven using MATLAB environment.

Fig. 5 depicts the I-V and P-V characteristics of the LG ELECTRONICS INC. LG400N2W-V5 PV panel under varied temperatures and solar radiation. Elevated temperatures reduce maximum power and open-circuit voltage, whereas increased radiation boosts short-circuit current, enhancing maximum output power. Temperature minimally affects current at MPP, but radiation substantially increases it.

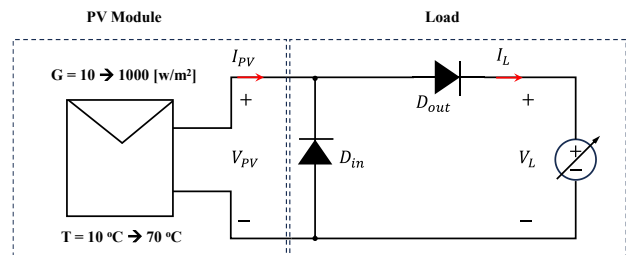
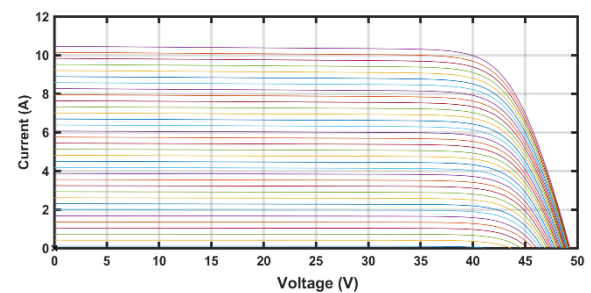


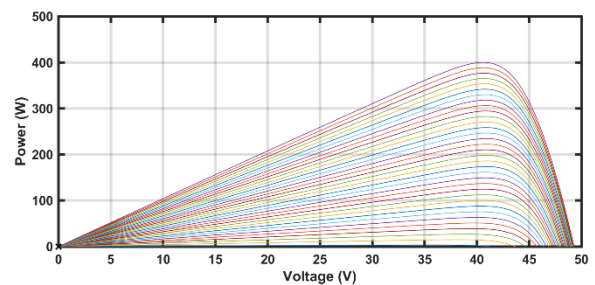
Fig. 4. Circuit model for data collection from the LG400N2W-V5 PV module used for training process.

Table 1: LG400N2W-V5 PV module parameters.

Parameter	Value	Unit
V_{OC}	49.3	[V]
I_{SC}	10.47	[A]
P_{MPP}	400.316	[W]
V_{MPP}	40.6	[V]
I_{MPP}	9.86	[A]



(a)



(b)

Fig. 5. LG400N2W-V5 PV panel characteristics under different radiation levels: (a) I-V and (b) P-V curves.

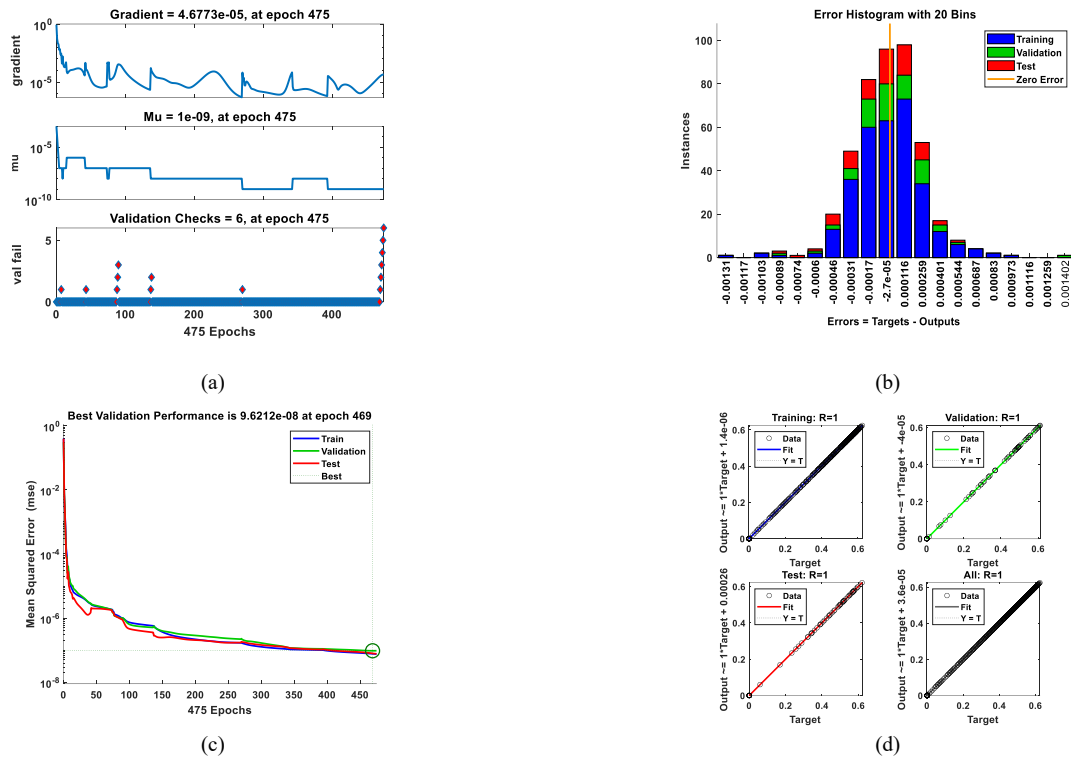


Fig. 6. The training process of the proposed ANN-based MPPT technique: (a) Training iterations; (b) Error histogram; (c) Mean square error (MSE) during iterative process; and (d) Regression ratio.

2) Network training and validation

The data stored from the pre-processing stage is sent to the ANN-fit MATLAB toolbox. Solar radiation and temperature are used as the input signals of the ANN tool

while the duty cycle of the semiconductor switch is used as the output of the ANN for ease of implementation.

The Levenberg-Marquardt algorithm with 20 layers with 70% of training data, 15% of validation data, and 15% of test data is used for the training process. The training process ended up with 475 epochs as seen in Fig. 6 (a) while the minimum error histogram is achieved to $-2.7e^{-5}$ as shown in

Fig. 6 (b). Fig. 6 (c) shows the Mean Square Error (MSE) reached its best validation performance at epoch 469 with $9.6212e^{-8}$ having the highest regression factor ($R=1$) as seen in Fig. 6 (d).

3) Implementation of the proposed ANN-MPPT technique

The final step involves exporting the trained ANN model to MATLAB-Simulink for simulation and hardware validation. The exported ANN fitting function is compatible with the MATLAB Embedded Coder toolbox, suitable for implementation on common microcontrollers. A direct implementation involves measuring solar radiation and PV panel temperature (refer to Fig. 7). The ANN function correlates sensed signals to trained data from I-V and P-V curves, generating the output duty cycle for the converter controller via Pulse-Width Modulator (PWM).

V. RESULTS AND DISCUSSION

Evaluating MPPT techniques in diverse weather conditions is crucial for PV systems. This study assesses the proposed ANN-MPPT's performance in optimizing energy

extraction under varying weather scenarios. Additionally, it includes a simulation comparison with conventional P&O technique.

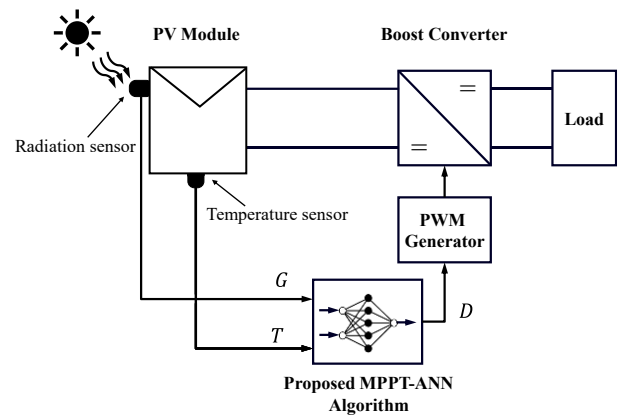


Fig. 7. Block diagram of the proposed ANN-MPPT technique.

In the first scenario, radiation transitions from 1000 W/m^2 to 400 W/m^2 , then to 800 W/m^2 at $t=0\text{s}$, $t=1\text{s}$, and $t=2\text{s}$, respectively, with constant temperature (30°C). As shown in Fig. 8, ANN techniques with convergence speeds of 0.075 s in zone 1, 0.025 s in zone 2, and 0.04 s in zone 3 (refer to Fig. 8 (a)). P&O displays the poorest performance, with the longest convergence speeds of 0.28 s in zone 1, 0.451 s in zone 2, and 0.364 s in zone 3. MPP values are 395 W at 1000 W/m^2 , 159 W at 400 W/m^2 , and 317 W at 600 W/m^2 . Corresponding duty cycles are illustrated in Fig. 8 (b).

In the second scenario, radiation constant 1000 W/m^2 at $t=0\text{s}$, $t=1\text{s}$, and $t=2\text{s}$, with a varied temperature from 50°C to 30°C then to 40°C respectively. As shown in Fig. 9, ANN techniques with convergence speeds of 0.073 s in zone 1, 0.01 s in zone 2, and 0.14 s in zone 3 (refer to Fig. 9 (a)). P&O

displays the poorest performance, with the longest convergence speeds of 0.3 s in zone 1, 0.21 s in zone 2, and 0.145 s in zone 3. MPP values are 367 W at 50°C, 394 W at 30°C, and 380 W at 40°C. Corresponding duty cycles are illustrated in Fig. 9 (b).

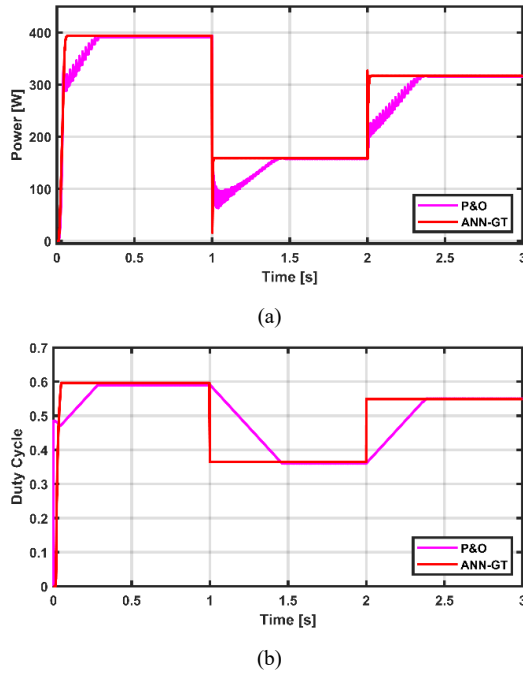


Fig. 8. Simulation results of the MPPT methods under investigation with radiation change (1000 → 400 → 800 W/m²) and constant temperature (30°C): (a) Output power and (b) Duty cycle.

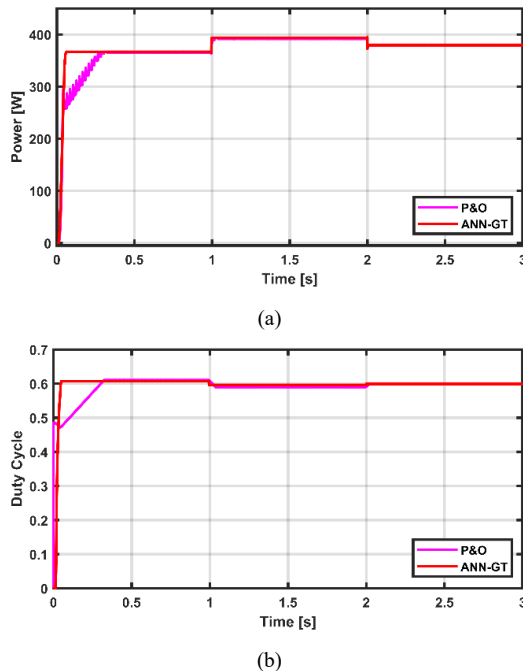


Fig. 9. Simulation results of the MPPT methods under investigation with temperature change (50 → 30 → 40 °C) and constant radiation (1000 W/m²): (a) Output power and (b) Duty cycle.

In the third scenario, radiation transitions from 1000 W/m² to 400 W/m², then to 800 W/m² at $t=0$ s, $t=1$ s, and $t=2$ s, with a

varied temperature from 50°C to 30°C then to 40°C respectively. As shown in Fig. 10, ANN techniques with convergence speeds of 0.073 s in zone 1, 0.02 s in zone 2, and 0.04 s in zone 3 (refer to Fig. 10 (a)). P&O displays the poorest performance, with the longest convergence speeds of 0.31 s in zone 1, 0.49 s in zone 2, and 0.38 s in zone 3. MPP values are 367 W at 1000 W/m², 159 W at 400 W/m², and 306.2 W at 600 W/m². Corresponding duty cycles are illustrated in Fig. 10 (b).

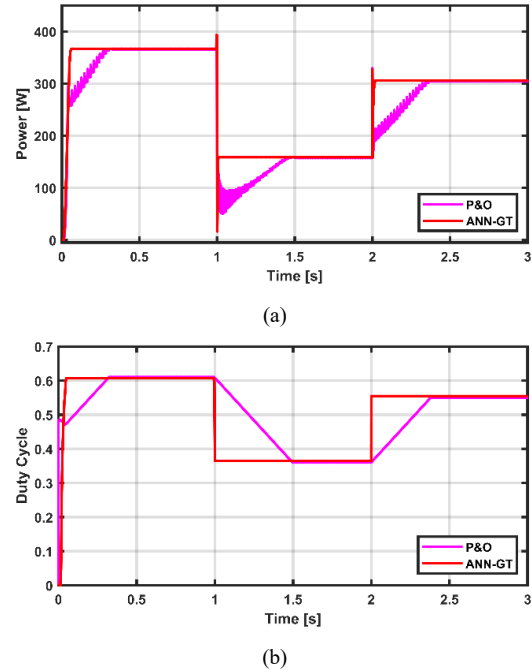


Fig. 10. Simulation results of the MPPT methods under investigation with radiation change (1000 → 400 → 800 W/m²) and temperature change (50 → 30 → 40 °C): (a) Output power and (b) Duty cycle.

VI. CONCLUSION

This paper presents the application of one of the common tools used in machine learning, i.e. artificial neural networks (ANNs) in PV systems. The proposed method copes with the limitations of the conventional maximum power tracking techniques such as the well-known Perturb & Observe method. These limitations include the slow peak tracking capability, low dynamics, and oscillating behavior around the peak point of the P-V curve.

The proposed ANN-MPPT method involves the Levenberg-Marquardt training algorithm to train and handle the non-linear correlations between the input variables (radiation and temperature) and the output variable (dc-dc converter duty cycle) of the PV system. The proposed ANN-MPPT efficiently models the complex connections between environmental parameters and the optimal operating point for MPP tracking, even in dynamic changing weather conditions with noisy measurements or insufficient dataset.

Finally, the proposed ANN-MPPT technique in comparison with the P&O method have been simulated and evaluated under identical weather conditions. The proposed technique showed a superior and robust performance with the highest MPP tracking capabilities compared to P&O method.

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